

TECHNISCHE UNIVERSITÄT BERLIN

Faculty IV – Electrical Engineering and Computer Science

Chair of Machine Learning and Security

MASTER'S THESIS

Official Thesis Title

BY

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Student ID: 000000

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Supervisor:	Your Supervisor

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Eigenständigkeits- erklärung

Hiermit versichere ich, dass ich die vorliegende Arbeit ohne Hilfe Dritter und ausschließlich unter Verwendung der aufgeführten Quellen und Hilfsmittel angefertigt habe. Alle Stellen die den benutzten Quellen und Hilfsmitteln wörtlich oder inhaltlich entnommen sind, habe ich als solche kenntlich gemacht.

Sofern KI-Tools verwendet wurden, habe ich Produktnamen, Hersteller, die jeweils verwendete Softwareversion und die Art der Nutzung (z. B. sprachliche Überprüfung und Verbesserung der Texte, systematische Recherche) benannt. Diese Angaben stehen in Abschnitt 4.4. Ich verantworte die Auswahl, die Übernahme und sämtliche Ergebnisse des von mir verwendeten KI-generierten Outputs vollumfänglich selbst.

Die Satzung zur Sicherung guter wissenschaftlicher Praxis an der TU Berlin vom 8. März 2017.

https://www.static.tu.berlin/fileadmin/www/10002457/K3-AMBI/Amtsblatt_2023/Amtliches_Mitteilungsblatt_Nr._16_vom_30.05.2023.pdf
habe ich zur Kenntnis genommen.

Ich erkläre weiterhin, dass ich die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegt habe.

Berlin, den 14. September 2026

Your Full Name

Zusammenfassung

Diese Zusammenfassung ist ein Platzhalter und zeigt, dass deutscher Text mit Umlauten und Silbentrennung funktioniert. Sie beschreibt in wenigen Sätzen das Problem, den Ansatz und das wichtigste Ergebnis der Arbeit. Ein Graph Neural Network (**GNN**) liest dabei die Struktur der Daten, und ein Large Language Model (**LLM**) erzeugt die Merkmale der Knoten. Die Kennzahlen stehen im deutschen Format, etwa ein Makro-F1 von 0,909 gegenüber 0,501 im Vergleichsfall. Der letzte Satz nennt, was die Arbeit zeigt, ohne neue Begriffe einzuführen.

Abstract

This abstract is a placeholder, written as one sentence per line so that a diff stays readable. It states the problem, the approach, and the result, and it repeats no sentence of the introduction verbatim. A **GNN** reads the structure of the data, and an **LLM** produces the node features. Numbers appear as they do in the chapters, a macro F1 of 0.909 against 0.501 for the baseline. The closing sentence says what the thesis shows.

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List of Acronyms

GNN Graph Neural Network

LLM Large Language Model

1 Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. A citation looks like this [MM23], and a citation with the author in the sentence reads Doe and Roe [DR24] instead. Acronyms expand on first use, so an LLM prints in full here and as LLM from the second use on. Cross-references go through cleveref, which supplies the word: Chapter 2 is a chapter, Section 4.1 a section, Table 1.1 a table.

1.1 Motivation

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium.¹ Totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo.

1.2 Research Question

Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt.

How does one phrase a research question so that an experiment can answer it?

Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit.

1.3 Contributions

1. **The first contribution.** Sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem (Chapter 4).

Table 1.1: A margin table. It sits in the 41.4mm margin column, so keep it to three or four narrow columns.

Variant	F1	Rec
baseline	0.501	0.206
proposed	0.909	0.847

1: A sidenote lands in the margin beside the line that calls it, numbered like a footnote.

2. **The second contribution.** Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam (Chapter 5).
3. **The third contribution.** Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur (Appendix A).

1.4 Thesis Outline

Chapter 2 reviews the literature, Chapter 3 describes the corpus, Chapter 4 defines the task and the evaluation protocol, Chapter 5 reports the experiments, Chapter 6 interprets them, and Chapter 7 closes with the findings and the open questions.

2 Background and Related Work

At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti. This chapter shows the citation styles, a margin figure, and an equation.

2.1 Foundations

Et harum quidem rerum facilis est et expedita distinctio, nam libero tempore, cum soluta nobis est eligendi optio.

Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae (Figure 2.1). An inline equation such as $f(x) = \sigma(Wx + b)$ sets in the text, whereas a numbered one stands on its own:

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad z \in \mathbb{R}. \quad (2.1)$$

Equation (2.1) is referenced the same way as everything else.

2.2 Related Work

Itaque earum rerum hic tenetur a sapiente delectus, ut aut reiciendis voluptatibus maiores alias consequatur aut perferendis doloribus asperiores repellat [Doe25; Exa26]. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus [DR24].

2.3 Research Gap

Omnis voluptas assumenda est, omnis dolor repellendus, and the gap this thesis addresses follows from the two paragraphs above.

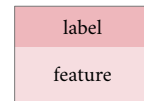


Figure 2.1: A margin figure, drawn with TikZ so it inherits the thesis colours. It is centred on the line that introduces it.

3 Dataset and Corpus Engineering

Section 3.1 describes where the samples come from and how they are labelled, Section 3.2 the pipeline that turns them into the representation the models read, and Section 3.3 reports what the corpus contains.

3.1 The Corpus

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua [Exa26]. State where each sample comes from, how it was labelled, and which decisions a later chapter depends on.

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat (Table 3.1).

Table 3.1: What the corpus holds, per class.

Class	Samples
negative	502
positive	213
total	715

3.2 From Raw Data to the Model's Input

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.



Figure 3.1: The pipeline from a raw sample to the stored representation. The accent colour marks the artifact the later chapters read from.

Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum (Figure 3.1).

3.3 What the Corpus Contains

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam. Report the counts a reader needs to judge the results, and give every one of them a source comment.

4 | Methodology

This chapter shows a full-width figure, a normal table, and a code listing.

4.1 Task and Protocol

Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur. Scores are reported as macro F1 and recall, with the Area Under the ROC Curve as context.



Figure 4.1: A figure at text width. Draw figures with TikZ so they inherit the thesis fonts and colours, which keeps them sharp at any zoom.

Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur (Figure 4.1).

4.2 Models and Training

Table 4.1: A normal table. Keep headers as plain text, since an acronym macro would render them as a coloured link.

Setting	Value
Hidden width	256
Layers	4
Learning rate	10^{-3}
Seeds	42, 43, 44

The configuration of Table 4.1 stays fixed across the experiments, so the rows of Chapter 5 differ in one factor at a time.

4.3 Implementation

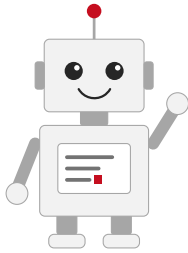
A listing sets in the monospace font of the template:

Listing 4.1: A code listing, numbered and referenced like a figure.

```
1 | def predict(graph, model):  
2 |     logits = model(graph.x, graph.edge_index)  
3 |     return logits.softmax(dim=1)[:, 1]
```

Listing 4.1 shows the call, and predict in the text uses the inline command.

4.4 Use of Generative AI Tools



Generative AI tools were used throughout this thesis and are named here with their purposes. The large language models that are the object of study are documented with their versions in Section 4.2 and are not meant here. Name each tool with its product name, its manufacturer, the version you used, and what you used it for, for example writing assistance, language editing, literature search, or code. Every design decision, every interpretation, and the final wording of every passage are my own, and I take full responsibility for the selection, adoption, and all results of the AI-generated output.

5 | Experiments and Results

Section 5.1 reports the main comparison under both fold sets, and Section 5.2 removes one component at a time.

5.1 Main Result

The proposed feature transfers, the baseline does not. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus, omnis voluptas assumenda est (Table 5.1). Both fold sets are reported side by side, so a single one cannot carry a claim on its own.

At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint.

Figure 5.1 carries the same comparison as a plot.

5.1.1 Reading

Itaque earum rerum hic tenetur a sapiente delectus, and the closing paragraph of a section states what the numbers above support without repeating them.

Table 5.1: A wide table. It spans the text block and the margin column, which fits a header of ten numeric columns; a normal table would not.

Feature	big-3					all-6				
	F1	AUC-ROC	AUC-PR	Prec	Rec	F1	AUC-ROC	AUC-PR	Prec	Rec
baseline	0.501	0.734	0.755	0.772	0.206	0.589	0.741	0.784	0.664	0.355
control	0.630	0.627	0.640	0.610	0.453	0.631	0.631	0.650	0.620	0.509
proposed	0.909	0.940	0.960	0.975	0.847	0.906	0.959	0.971	0.912	0.868

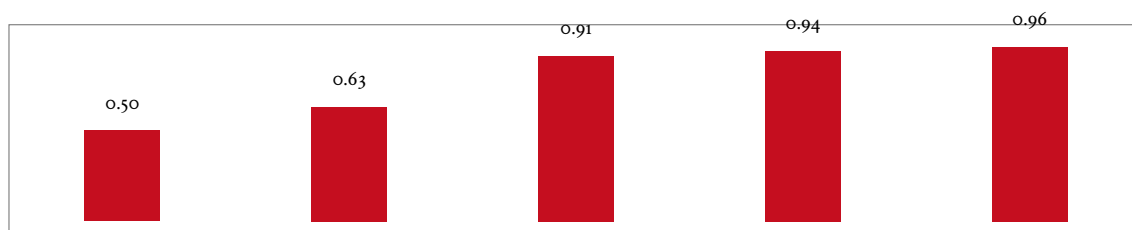


Figure 5.1: A wide figure, the same width as a wide table. Use it when a plot needs the extra 41.4 mm.

5.2 Ablation

Removing one component at a time isolates its contribution (Table 5.2).

Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae. The component that matters most is the one whose removal costs the most, which Table 5.2 reports against the full configuration of Table 5.1.

One factor at a time. Et harum quidem rerum facilis est et expedita distinctio, nam libero tempore, cum soluta nobis est eligendi optio.

5.2.1 Reading

Omnis voluptas assumenda est, omnis dolor repellendus, and this closing paragraph says which component the chapter attributes the result to.

Table 5.2: One factor at a time, beside the paragraph that reads it.

Variant	F1
full	0.909
– feature	0.501
– structure	0.630

6 | Discussion

Section 6.1 reads the main result against the baseline, Section 6.2 traces it to a mechanism, and Section 6.3 closes with the limitations and the threats to validity.

6.1 What the Numbers Support

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis (Table 5.1). A discussion interprets results that a previous chapter established, so it cites them rather than restating the table. The gap between 0.501 and 0.909 holds under every fold set, which is what makes it a finding instead of an artifact of one split.

6.2 Why It Happens

Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem [MM23].

A named step in the argument. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosam, nisi ut aliquid ex ea commodi consequatur. A paragraph heading carries one step of the argument and keeps a long section navigable.

6.3 Limitations and Threats to Validity

Every result rests on one corpus, one protocol, and three seeds, so each number carries the uncertainty of that setup. At vero eos et accusamus et iusto odio dignissimos ducimus qui blanditiis praesentium voluptatum deleniti atque corrupti quos dolores et quas molestias excepturi sint.

Construct validity. Et harum quidem rerum facilis est et expedita distinctio, and the measure may capture something adjacent to what the question asks about.

Internal validity. Nam libero tempore, cum soluta nobis est eligendi optio cumque nihil impedit quo minus id quod maxime placeat facere possimus.

External validity. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet ut et voluptates repudiandae sint et molestiae non recusandae.

7 Conclusion and Future Work

This chapter states the findings, the limitations, and what follows from them.

7.1 Findings

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam (Chapter 5). The proposed feature reaches 0.909 big-3 and 0.906 all-6 macro F1, against 0.501 and 0.589 for the baseline.

7.2 Future Work

Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt.

References

- [Doe25] J Doe. A Preprint, Cited the Way arXiv Entries Are. 2025. arXiv: 2501.00000 [cs.CR] (page 3).
- [DR24] J Doe and R Roe. A Conference Paper with a Reasonably Long Title. In: *Proceedings of the Example Conference on Applied Things*. 2024, pp. 101–117 (pages 1, 3).
- [Exa26] Example Organization. A Website, for the URL and urldate Fields. 2026. <https://example.org/a/page> (visited on 01/15/2026) (pages 3, 5).
- [MM23] E Musterfrau and M Mustermann. A Journal Article Used to Show the Citation Style. In: *Journal of Example Studies* 12(3):(2023), 45–62 (pages 1, 13).

A | Reproducibility

The appendices run at full width: the margin column is folded into the text block after `\appendix`, so tables and listings get 164.6 mm instead of 115 mm. A wide table is therefore unnecessary here, and a normal table already spans the block.

A.1 Configuration

Table A.1: An appendix table. It uses the full 164.6 mm, which is why the appendices carry no margin elements.

Field	Value	Note
Model	the model identifier	as published
Revision	the commit of the model repository	pin it, tags move
Precision	bfloat16, no quantization	
Seeds	42, 43, 44	three per configuration

Table [A.1](#) prints as "Table A.1", and Appendix [A.1](#) prints as "Appendix A.1", because `\crefalias` is set after `\appendix`.

B | Prompts and Listings

Long verbatim material belongs in an appendix, where the full width keeps lines from wrapping.

B.1 A Prompt

Listing B.1: A prompt, set verbatim so that every character is the one that was sent.

```
1 | You are given the source of one class.  
2 | Answer with JSON only, using this schema:  
3 |   {"malicious": true|false, "reason": "<one sentence>"}  
4 | Judge the code, not the class name.
```

B.2 An Output

Listing B.2: The model's answer to the prompt of Listing B.1.

```
1 | {"malicious": false, "reason": "Reads a bundled asset and decodes it."}
```